

Yan Yan



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ACADEMIC REFERENCE

Professor Salvador Pané, Multi-Scale Robotics Lab, ETH Zurich vidalp@ethz.ch
Dr. Luis Deán-Ben, Group Leader, Multiscale Functional and Molecular Imaging Lab, ETH xl.deanben@pharma.uzh.ch

EDUCATION

M.Sc., Biomedical Engineering, University Basel, Switzerland 09.2023-06.2026
Current GPA: 5.3/6

Medical Imaging and Medical Image Processing, Model-Based Signal Processing Optimization Methods, Advanced Methods in Medical Image Analysis, Principles of Medical Imaging, Introduction to LTI Systems and Control, Mechanics, Modelling and Simulation (COMSOL), Materials Science and Biomaterials, Electrical Engineering and Electronics Basics

B.Sc., Dental Technology, Sichuan University, West China School of Stomatology, China 09.2019-06.2023
GPA: 88.24/100

Main Courses: Mathematical Analysis, Advanced Algebra, Analytic Geometry, Medical Statistics, Medical Physics, Medical Biology, Histology and Pathology, Anatomy and Physiology, Organic Chemistry and Experiments, Dental Materials, Oral and Maxillofacial Imaging Diagnostics

RESEARCH INTERNSHIP

Multiscale Functional and Molecular Imaging Lab, ETH Zurich, Switzerland (Master Thesis) 07.2025-Present
Advisor: Dr. Luis Deán-Ben

- Performed optoacoustic image reconstruction using homogeneous and heterogeneous back-projection as well as LSQR-based heterogeneous models to reconstruct in vivo mouse abdomen images
- Implemented and compared vesselness enhancement filters (Frangi, Jerman, and others) for post-processing
- Developed an unrolled, physics-guided deep learning framework that integrates CNN-based image enhancement with model-based reconstruction, where the CNN acts as a self-learned regularizer
- Conducted systematic comparison and evaluation of reconstruction performance across different algorithm

Multi-Scale Robotics Lab, ETH Zurich, Switzerland (Semester Thesis) 10.2024-07.2025
Advisor: Huimin Chen, Dr. Semih Sevim, (under supervision of Prof. Salvador Pané)
Grade: 5.75/6

- Determined and optimized PEGDA/PNIPAM hydrogel formulations for DLP-based 3D printing
- Fabricated PDMS-acrylic pool chambers for hydrogel photocuring via PDMS spin-coating and oxygen-plasma bonding
- Designed and 3D-printed bilayer hydrogel cuff electrodes integrating PNIPAM and PEGDA
- Controlled curvature deformation under thermal stimulation through structural design optimization
- Designed and 3D-printed flow chambers to evaluate cuff-electrode adhesion under fluid conditions
- Utilized CAD modeling in Onshape, UV polymerization setup, and optical imaging techniques for characterization

INDUSTRIAL INTERNSHIP

Chengdu DT Denture Technology Development Co., Ltd., China 02.2023-06.2023
• Learned the management mode of the company in the process of denture production
• Fabricated the dental prosthesis via CAD/CAM

West China Hospital of Stomatology, Sichuan University, National Center of Stomatology 07.2022-06.2023
• Assisted clinical doctors to complete patients' diagnoses, information collection and medical scanning
• Scanned the digital dentition and dental models for further landmarking and analysis
• Fabricated the dental prosthesis via CAD/CAM

Chengdu Boltzmann Zhibei Intelligence Technology Co., Ltd., China 06.2021-09.2021
• Referred to the literature about the CBCT images quality and airway analysis on lateral cephalogram
• Processed the CBCT images via the ITK-SNAP software to segment and mark the dentition from the images
• Participated in a research-oriented project related to cervical vertebral maturation analysis

PUBLICATION

Jiang F, Abdulqader AA, Yan Y, Cheng F, Xiang T, Yu J, Li J, Qiu Y, Chen X. Deep learning based quantitative cervical vertebral maturation analysis. Head Face Med. 2025 Mar 26;21(1):20. doi: 10.1186/s13005-025-00498-6. PMID: 40140932; PMCID: PMC11938625.

- Contributed to manuscript preparation and assisted in data summary and result interpretation

SKILLS

Language: Mandarin (native), English (fluent), German (Basic)

Programming: Python, MATLAB, LaTeX, R, HTML/CSS

Hobby: Reading, Traveling, Swimming, Wandering